St. Francis Institute of Technology, Mumbai-400 103
Department of Information Technology

A.Y. 2025-2026
Class: BE-ITA/B, Semester: VIII
Subject: BlockChain Lab

Experiment – 3

1. **Aim:** To implement smart contract using Remix IDE and deploy it using MetaMask.
2. **Objective:** To …
   - explore working of Local Ethereum network.
   - explore working of MetaMask wallet
   - test a smart contract by using Remix IDE
3. **Lab outcome:** After performing the experiment, the students will be able to **implement** smart contracts in Ethereum using different development frameworks (PO3, PSO2, BL3)
4. **Prerequisite:**
   - Fundamental knowledge of blockchain
   - Knowledge of the Ethereum platform and Remix IDE
   - Familiarity with the Solidity programming language
5. **Requirements:** The following are the requirements –
   Remix IDE, MetaMask wallet etc.
6. **Pre-Experiment Theory:**

What is MetaMask?
MetaMask is a browser plugin that serves as a cryptocurrency wallet and enables users to store Ether and other ERC-20 tokens. It also enables users to access the Web 3 ecosystem of dapps. It relies on browser integration and good design to serve as one of the main gateways to the world of Web3, decentralized finance (DeFi) and NFTs.

7. **Laboratory Exercise Steps to be implemented.**

To Follow the procedure given below to build smart contract in Remix IDE

1. Download and install the official MetaMask extension/ plugin.
2. Open Remix IDE in Google Chrome.
3. Write smart contract by creating new file under contract folder, with .sol extension.
4. Write your contract code using solidity language.
5. Click on compiler icon, choose compiler version or keep default and click on compile button.
6. For deployment…
   In Remix IDE choose environment Injected Provider- MetaMask and then deploy the said contract.
7. Under deployed contract, get the output of your contract.

Program Code

Write a smart contract StudentMarks.sol to create student database with attributes, id, name, dept, sub1marks, sub2Marks etc. and deploy it using Remix and MetaMask.

8. Post Experimental Exercise-

A. Questions:

- List down the details of different MetaMask networks
- Create any one game using smart contract and deploy it.

B. Results/Observations/Program output:

Present the program input/output results if any and comment on the same.

C. Conclusion:

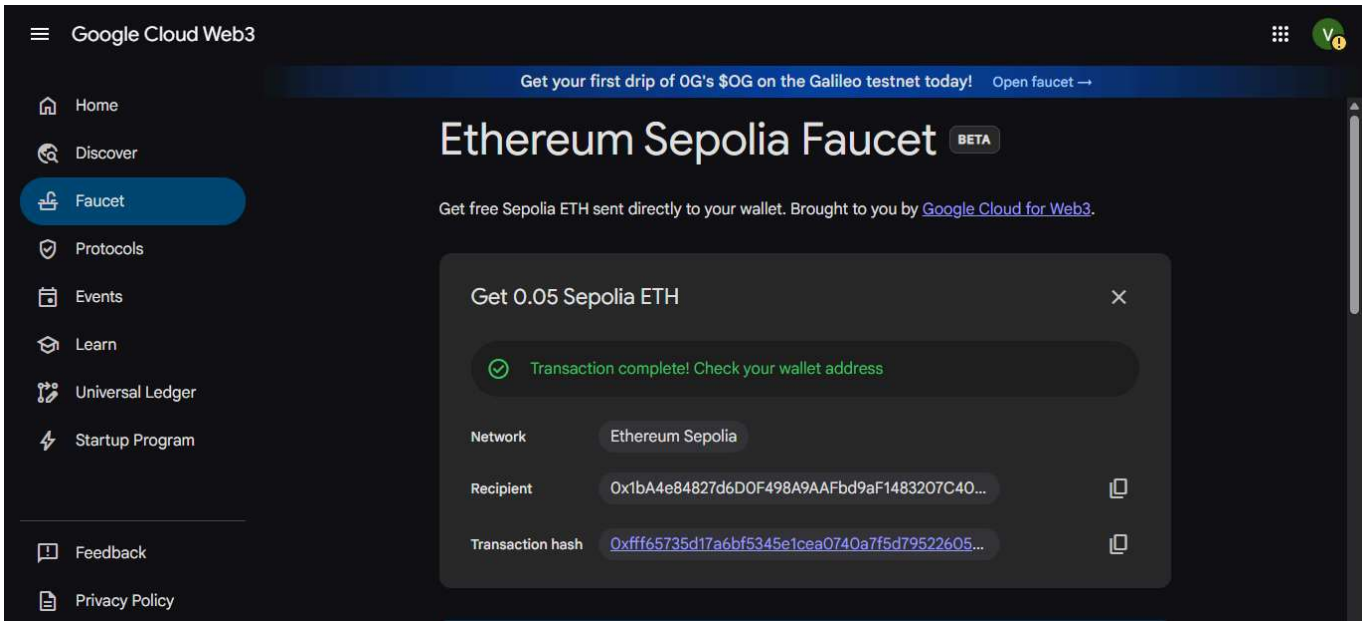
- Write what was performed in the experiment.
- Write which tools you used to perform the experiment
- Write what you inferred from the output obtained

D. References:

https://www.geeksforgeeks.org/how-to-use-metamask-to-deploy-a- smart-contract-in-solidity-blockchain/

Mastering Ethereum, Building Smart Contract and Dapps, Andreas M. Antonopoulos Dr. Gavin Wood, O'reilly

LABORATORY EXERCISE



Write a smart contract StudentMarks.sol to create student database with attributes, id, name, dept, sub1marks, sub2Marks etc. and deploy it using Remix and MetaMask.

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract StudentMarks {

    // Structure to store student details
    struct Student {
        uint id;
        string name;
        string department;
        uint sub1Marks;
        uint sub2Marks;
    }

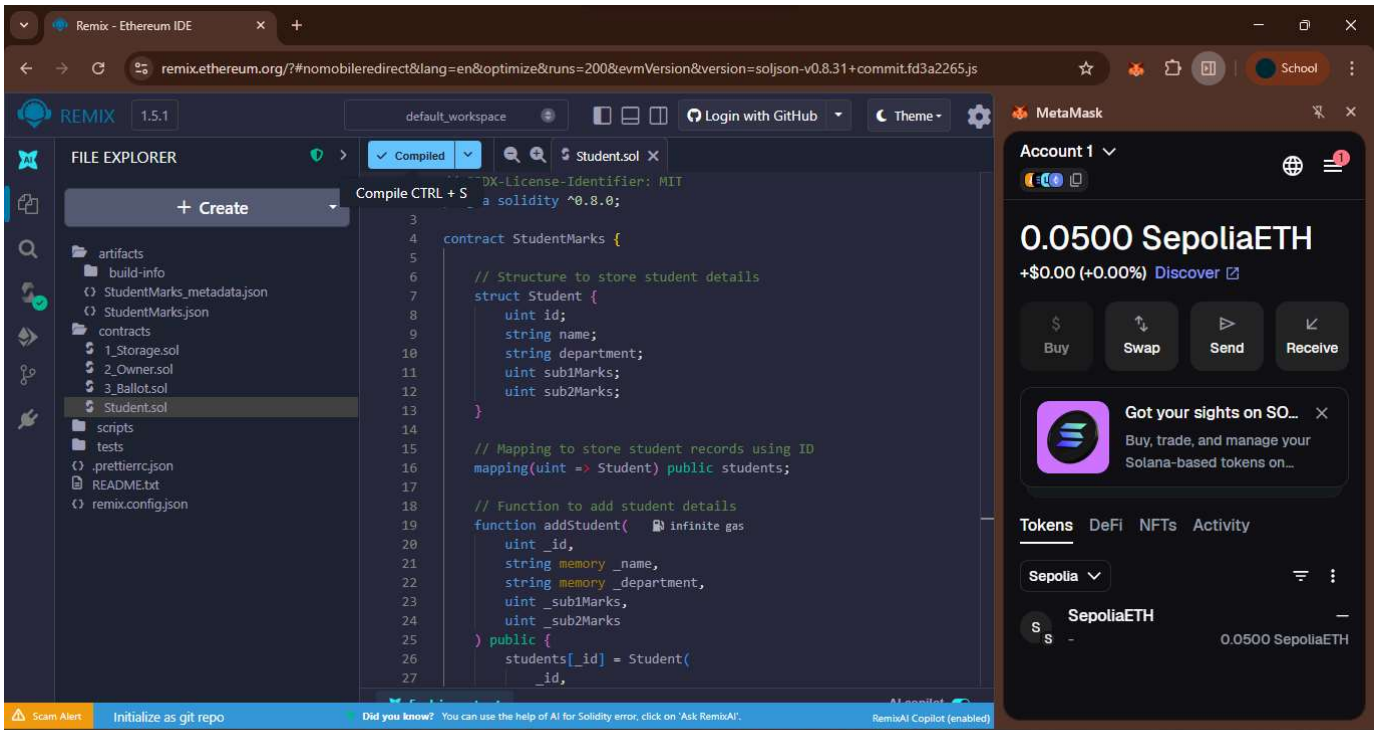
    // Mapping to store student records using ID
    mapping(uint => Student) public students;

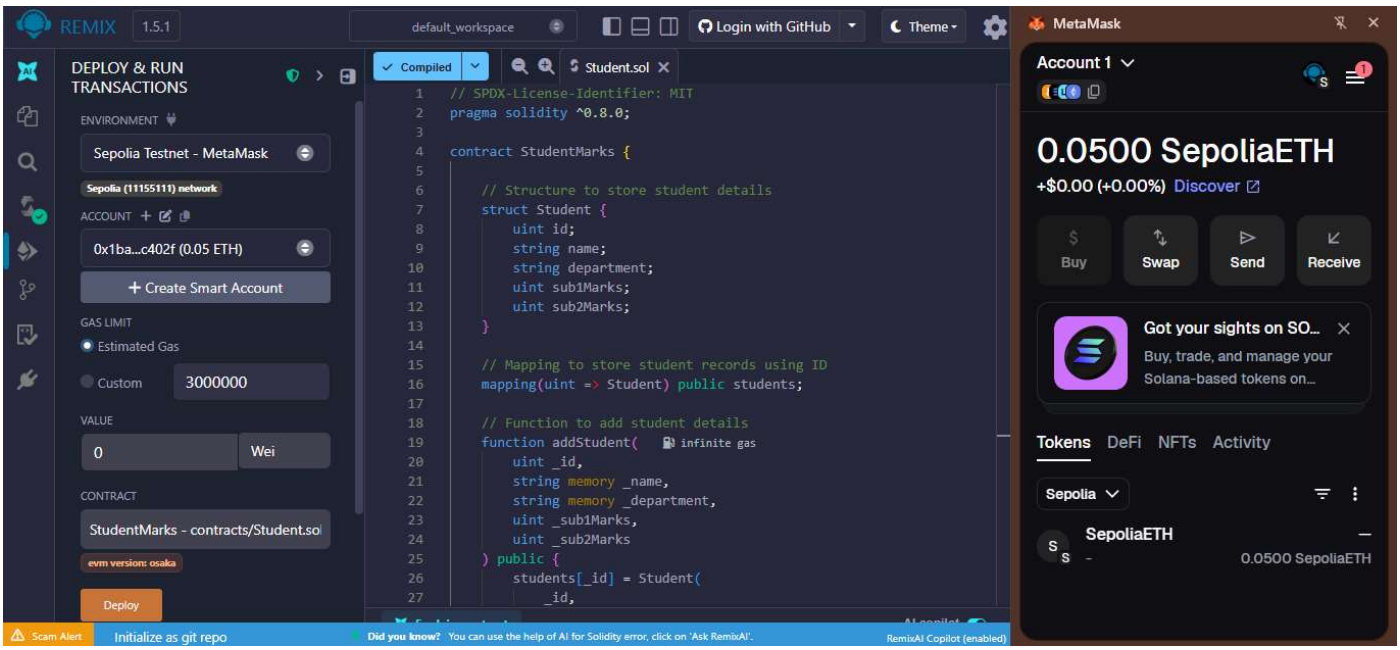
    // Function to add student details
    function addStudent(
        uint _id,
        string memory _name,
        string memory _department,
        uint _sub1Marks,
        uint _sub2Marks
    ) public {

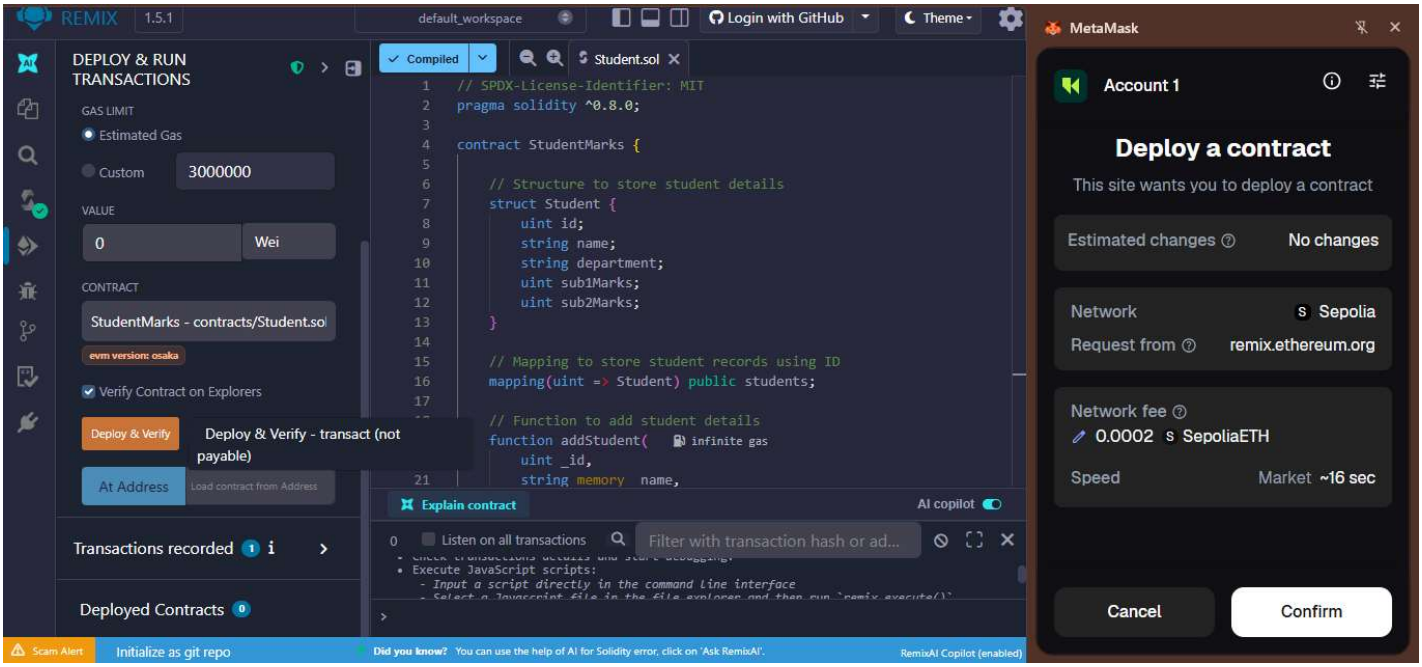
        students[_id] = Student(
            id
```

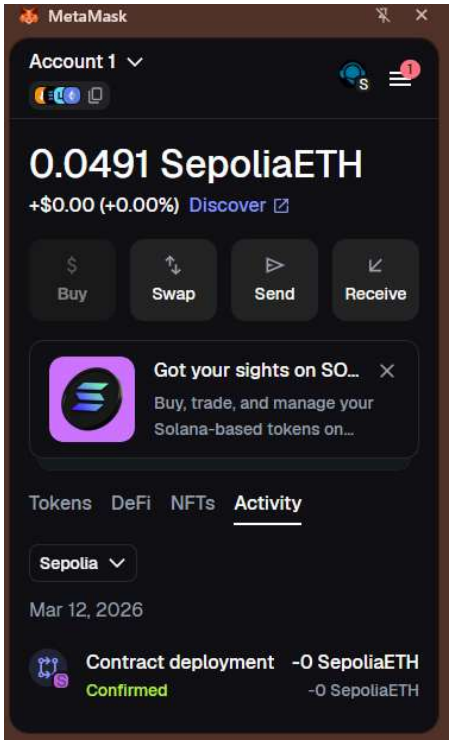
```
// Function to get student details
    function getStudent(uint _id)
        public
        view
        returns (
            uint,
            string memory,
            string memory,
            uint,
            uint
        )
    {
        Student memory s = students[_id];

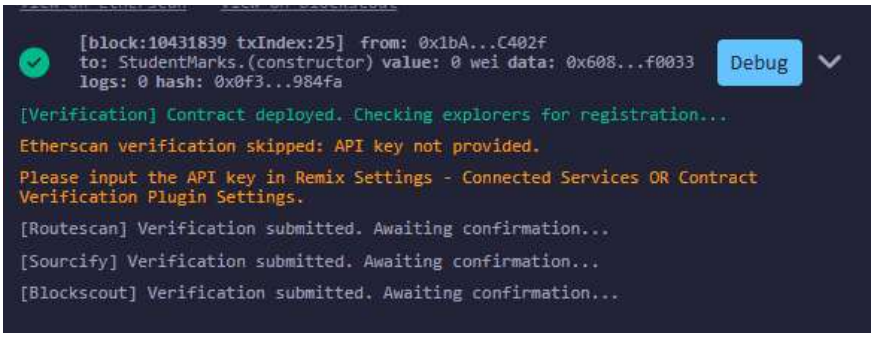
        return (
            s.id,
            s.name,
            s.department,
            s.sub1Marks,
            s.sub2Marks
        );
    }
}
```

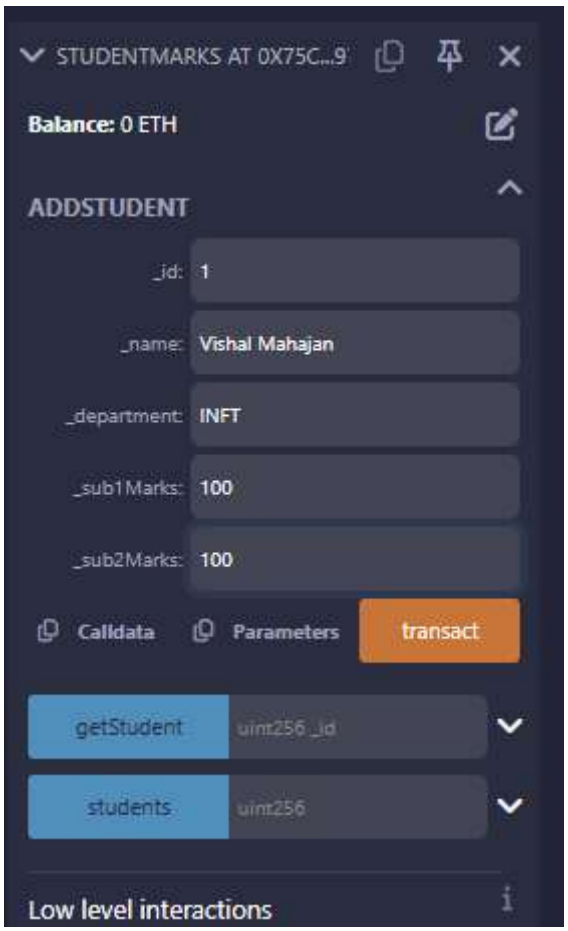

Remix - Ethereum IDE
remix.ethereum.org/?#nomobileredirect&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.31+commit.fd3a2265.js
REMIX 1.5.1
default_workspace
Login with GitHub
Theme
FILE EXPLORER
+ Create
artifacts
build-info
StudentMarks_metadata.json
StudentMarks.json
contracts
1_Storage.sol
2_Owner.sol
3_Ballot.sol
Student.sol
scripts
tests
.prettierrc.json
README.txt
remix.config.json
Compiled
Student.sol
Compile CTRL + S
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract StudentMarks {
// Structure to store student details
struct Student {
uint id;
string name;
string department;
uint sub1Marks;
uint sub2Marks;
}
// Mapping to store student records using ID
mapping(uint => Student) public students;
// Function to add student details
function addStudent(infinite gas
uint _id,
string memory _name,
string memory _department,
uint _sub1Marks,
uint _sub2Marks
) public {
students[_id] = Student(
_id,
Scam Alert
Initialize as git repo
Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.
RemixAI Copilot (enabled)
MetaMask
Account 1
0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Got your sights on SO...
Buy, trade, and manage your Solana-based tokens on...
Tokens
DeFi
NFTs
Activity
Sepolia
SepoliaETH
0.0500 SepoliaETH

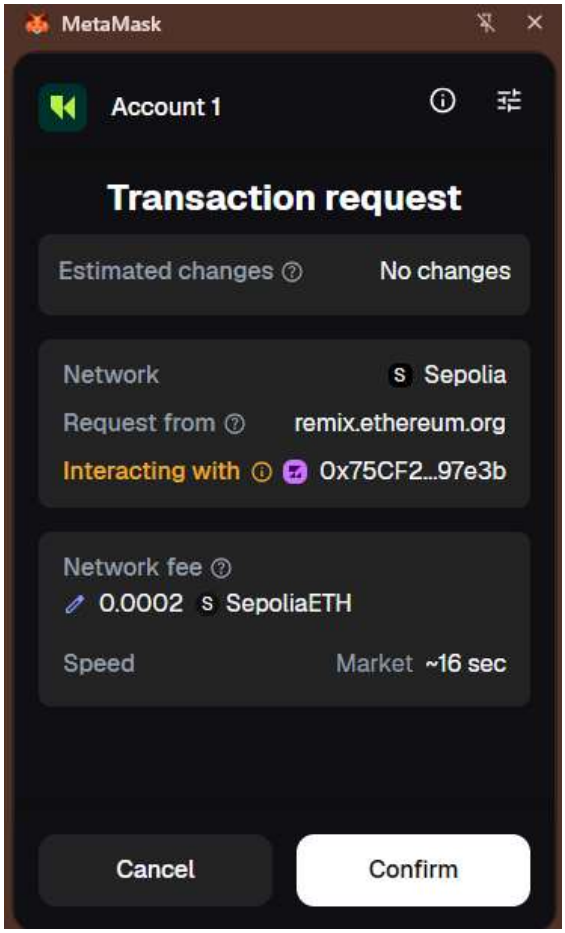

REMIX 1.5.1
default_workspace
Login with GitHub
Theme
DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Sepolia Testnet - MetaMask
Sepolia (11155111) network
ACCOUNT
0x1ba...c402f (0.05 ETH)
+ Create Smart Account
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
StudentMarks - contracts/Student.sol
evm version: osaka
Deploy
Compiled
Student.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract StudentMarks {
// Structure to store student details
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uint id;
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REMIX 1.5.1
default_workspace
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Theme
DEPLOY & RUN TRANSACTIONS
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
StudentMarks - contracts/Student.sol
evm version: osaka
Verify Contract on Explorers
Deploy & Verify
Deploy & Verify - transact (not payable)
At Address
Load contract from Address
Transactions recorded
Deployed Contracts
Compiled
Student.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract StudentMarks {
// Structure to store student details
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uint id;
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}
// Mapping to store student records using ID
mapping(uint => Student) public students;
// Function to add student details
function addStudent(
infinite gas
uint _id,
string memory name,
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
Execute JavaScript scripts:
Input a script directly in the command line interface
Scam Alert
Initialize as git repo
Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.
RemixAI Copilot (enabled)
MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Network fee
0.0002 SepoliaETH
Speed
Market ~16 sec
Cancel
Confirm


MetaMask
Account 1
0.0491 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Got your sights on SO...
Buy, trade, and manage your Solana-based tokens on...
Tokens
DeFi
NFTs
Activity
Sepolia
Mar 12, 2026
Contract deployment
-0 SepoliaETH
Confirmed
-0 SepoliaETH


[block:10431839 txIndex:25] from: 0x1bA...C402f
to: StudentMarks.(constructor) value: 0 wei data: 0x608...f0033
logs: 0 hash: 0x0f3...984fa
Debug
[Verification] Contract deployed. Checking explorers for registration...
Etherscan verification skipped: API key not provided.
Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.
[Routescan] Verification submitted. Awaiting confirmation...
[Sourcify] Verification submitted. Awaiting confirmation...
[Blockscout] Verification submitted. Awaiting confirmation...


STUDENTMARKS AT 0X75C...9
Balance: 0 ETH
ADDSTUDENT
_id: 1
_name: Vishal Mahajan
_department: INFT
_sub1Marks: 100
_sub2Marks: 100
Calldata
Parameters
transact
getStudent
uint256 _id
students
uint256
Low level interactions


MetaMask
Account 1
Transaction request
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Interacting with
0x75CF2...97e3b
Network fee
0.0002 SepoliaETH
Speed
Market ~16 sec
Cancel
Confirm


getStudent
1
0: uint256: 1
1: string: Vishal Mahajan
2: string: INFT
3: uint256: 100
4: uint256: 100
students
1
0: uint256: id 1
1: string: name Vishal Mahajan
2: string: department INFT
3: uint256: sub1Marks 100
4: uint256: sub2Marks 100

POST EXPERIMENT:

Create any one game using smart contract and deploy it.

CODE:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract GuessTheNumber {

    address public owner;
    uint private secretNumber;
    bool public gameActive;

    event CorrectGuess(address player);
    event WrongGuess(address player, uint
guessedNumber);

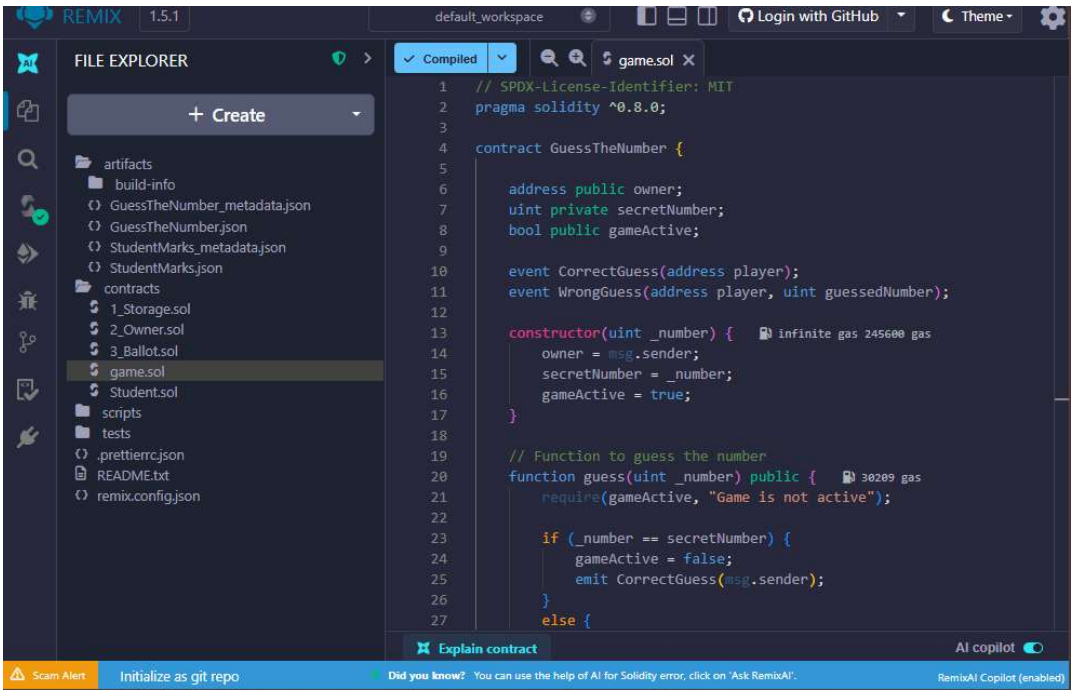
    constructor(uint _number) {
        owner = msg.sender;
        secretNumber = _number;
        gameActive = true;
    }
```

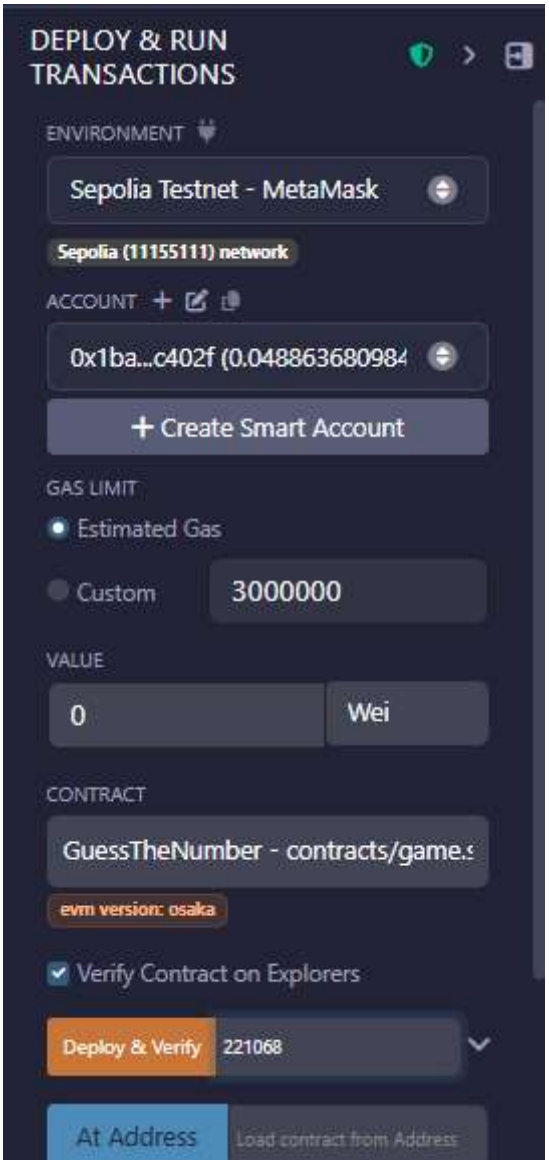
```
// Function to guess the number
    function guess(uint _number) public {
        require(gameActive, "Game is not active");

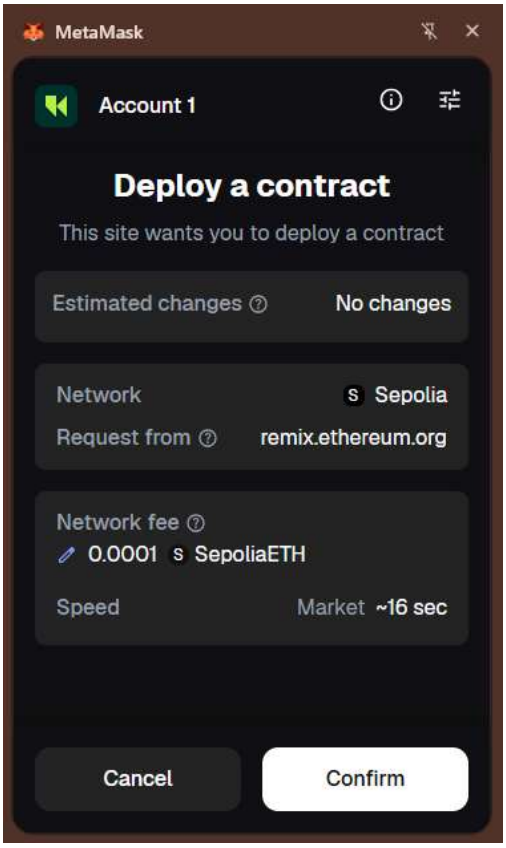
        if (_number == secretNumber) {
            gameActive = false;
            emit CorrectGuess(msg.sender);
        }
        else {
            emit WrongGuess(msg.sender,
_number);
        }
    }

    // Reset game (only owner)
    function resetGame(uint _newNumber) public
{
        require(msg.sender == owner, "Only owner
can reset");

        secretNumber = _newNumber;
        gameActive = true;
    }
}
```

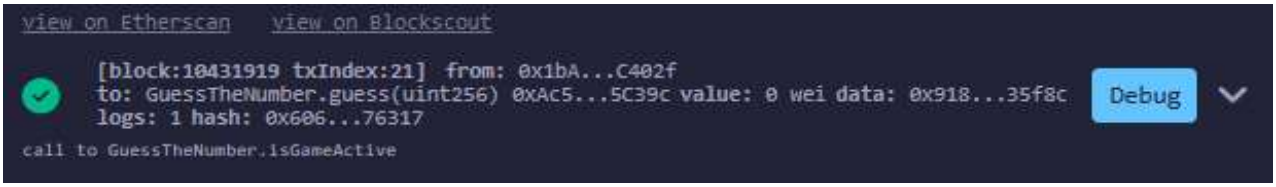



DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Sepolia Testnet - MetaMask
Sepolia (11155111) network
ACCOUNT
0x1ba...c402f (0.048863680984
+ Create Smart Account
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
GuessTheNumber - contracts/game.s
evm version: osaka
Verify Contract on Explorers
Deploy & Verify
221068
At Address
Load contract from Address


MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Network fee
0.0001 SepoliaETH
Speed
Market ~16 sec
Cancel
Confirm


[block:10431873 txIndex:36] from: 0x1bA...C402f to: GuessTheNumber.(constructor)
value: 0 wei data: 0x608...35f8c logs: 0 hash: 0xba3...24c28
Debug
[Verification] Contract deployed. Checking explorers for registration...
Etherscan verification skipped: API key not provided.
Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.
[Routescan] Verification submitted. Awaiting confirmation...
[Sourcify] Verification submitted. Awaiting confirmation...
[Blockscout] Verification submitted. Awaiting confirmation...
[Sourcify] Verification Successful! View Code
[Blockscout] Verification Successful! View Code


GUESSTHENUMBER AT 0XAC5
Balance: 0 ETH
guess
221068
resetGame
uint256 _newNumber
gameActive
isGameActive
0: bool: true
owner
0: address: 0x1bA4e84827d6D0F498
A9AAFbd9aF1483207C402f


view on Etherscan
view on Blockscout
[block:10431919 txIndex:21] from: 0x1bA...C402f
to: GuessTheNumber.guess(uint256) 0xAc5...5C39c value: 0 wei data: 0x918...35f8c
logs: 1 hash: 0x606...76317
Debug
call to GuessTheNumber.isGameActive